REAR AXLE - 216RBI (Continued)

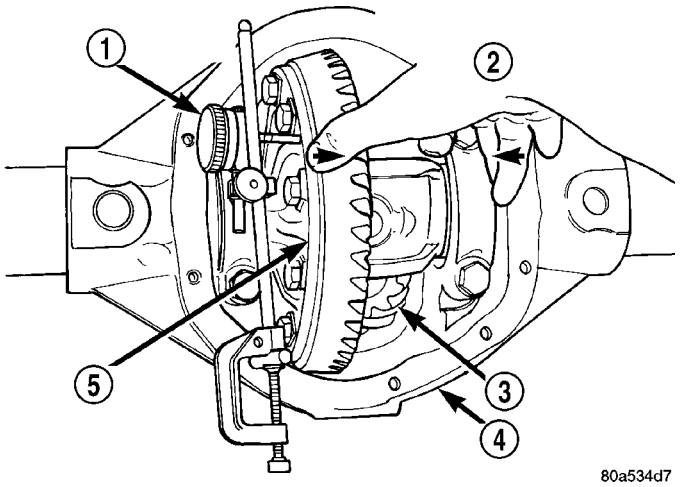


Fig. 18 ZERO DIAL INDICATOR

1 - DIAL INDICATOR FACE
2 - DIFFERENTIAL CASE TO PINION GEAR SIDE
3 - PINION GEAR
4 - DIFFERENTIAL HOUSING
5 - DIFFERENTIAL CASE

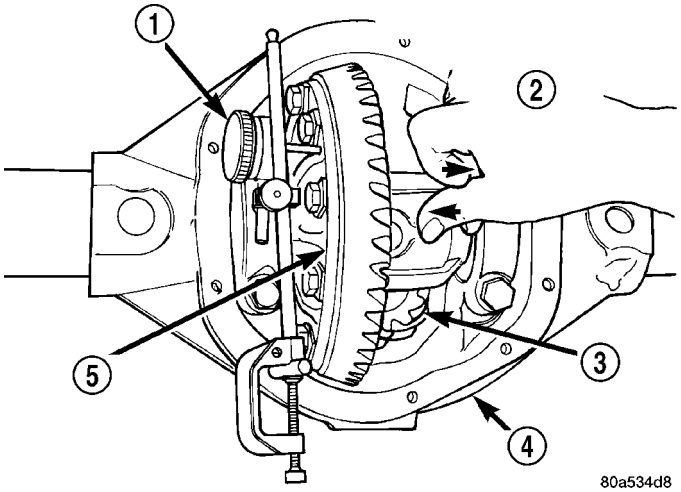


Fig. 19 RECORD DIAL INDICATOR READING

1 - DIAL INDICATOR
2 - DIFFERENTIAL CASE TO RING GEAR SIDE
3 - PINION GEAR
4 - DIFFERENTIAL HOUSING
5 - DIFFERENTIAL CASE

(24) Rotate dial indicator out of the way on pilot stud.

(25) Remove differential case and dummy bearings from the housing.

(26) Install side bearing shims on differential case hubs.

(27) Install new side bearing cones and cups on differential case.

(28) Install spreader W-129-B on the housing and spread axle opening enough to receive differential case.

(29) Install differential case in the housing.

(30) Install differential bearing caps loosely.

(31) Remove spreader from the housing.

(32) Tighten bearing caps bolts to 108 N·m (80 ft. lbs.).

(33) Rotate the differential case several times to seat the side bearings.

(34) Position the indicator plunger against a ring gear tooth (Fig. 20).

(35) Push and hold ring gear upward while not allowing the pinion gear to rotate and zero dial indicator.

(36) Push and hold ring gear downward while not allowing the pinion gear to rotate. Dial indicator reading should be between 0.12 mm - 0.20 mm (0.005 in. - 0.008 in.). If backlash is not within specifications transfer the necessary amount of shim thickness from one side of the housing to the other (Fig. 21).

(37) Verify differential case and ring gear runout by measuring ring to pinion gear backlash at eight locations around the ring gear. Readings should not vary more than 0.05 mm (0.002 in.). If readings vary more than specified, the ring gear or the differential case is defective.

After the proper backlash is achieved, perform Gear Contact Pattern Analysis procedure.

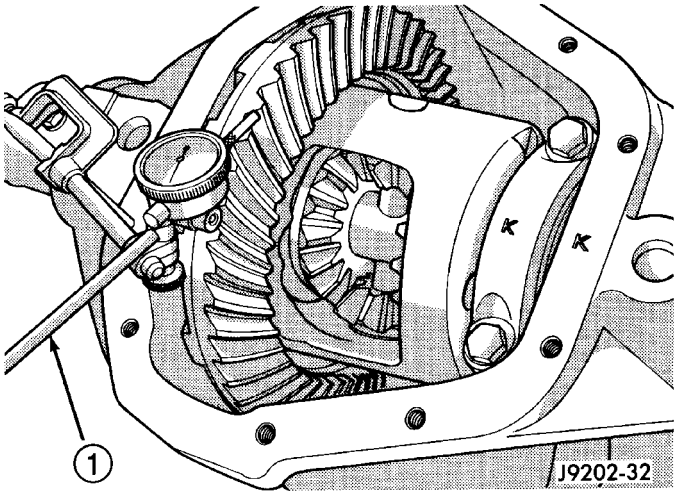


Fig. 20 RING GEAR BACKLASH MEASUREMENT

1 - DIAL INDICATOR

GEAR CONTACT PATTERN

The ring gear and pinion teeth contact patterns will show if the pinion depth is correct in the housing. It will also show if the ring gear backlash has been adjusted correctly. The backlash can be adjusted within specifications to achieve desired tooth contact patterns.

The TOP LAND of the gear tooth is the top surface of the tooth. The PROFILE of the gear tooth is the depth of the tooth. The TOE of the gear is the portion of the tooth surface at the end towards the center. The HEEL of the gear is the portion of the tooth